D Y PATIL AGRICULTURE & TECHNICAL UNIVERSITY TALSANDE

MCA SANDWICH

PROJECT NAME

HEART DISEASE ANALYSIS

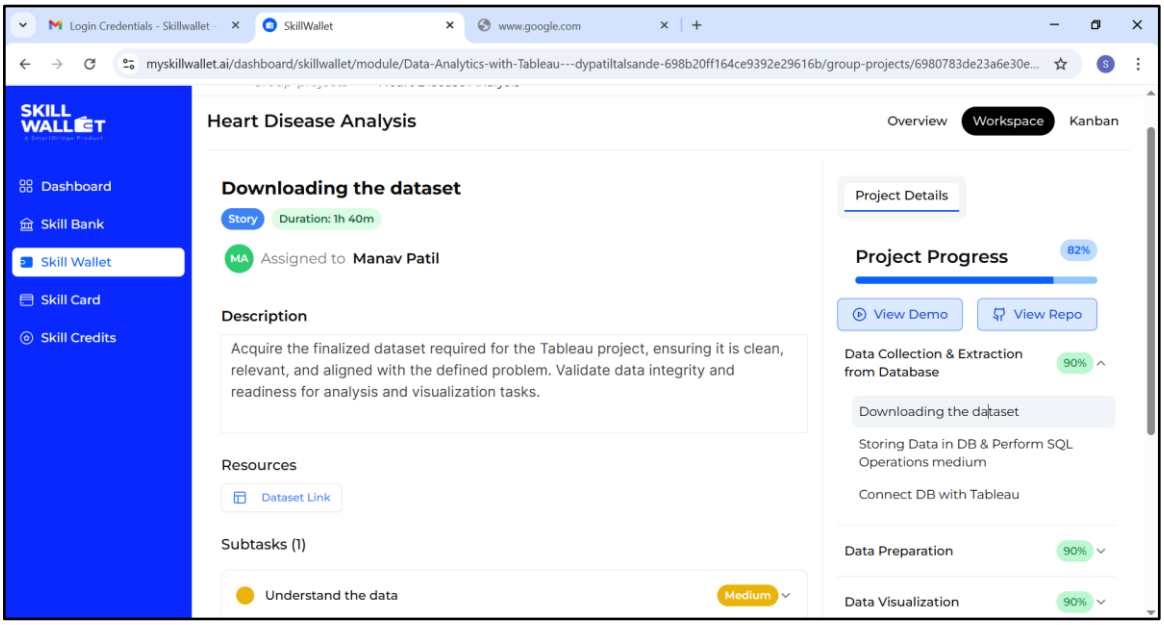
TEAM MEMBER

1) Manav Patil
2) Swaranjali Mohite
3) Shweta Patil
4) Tanaya Kagale

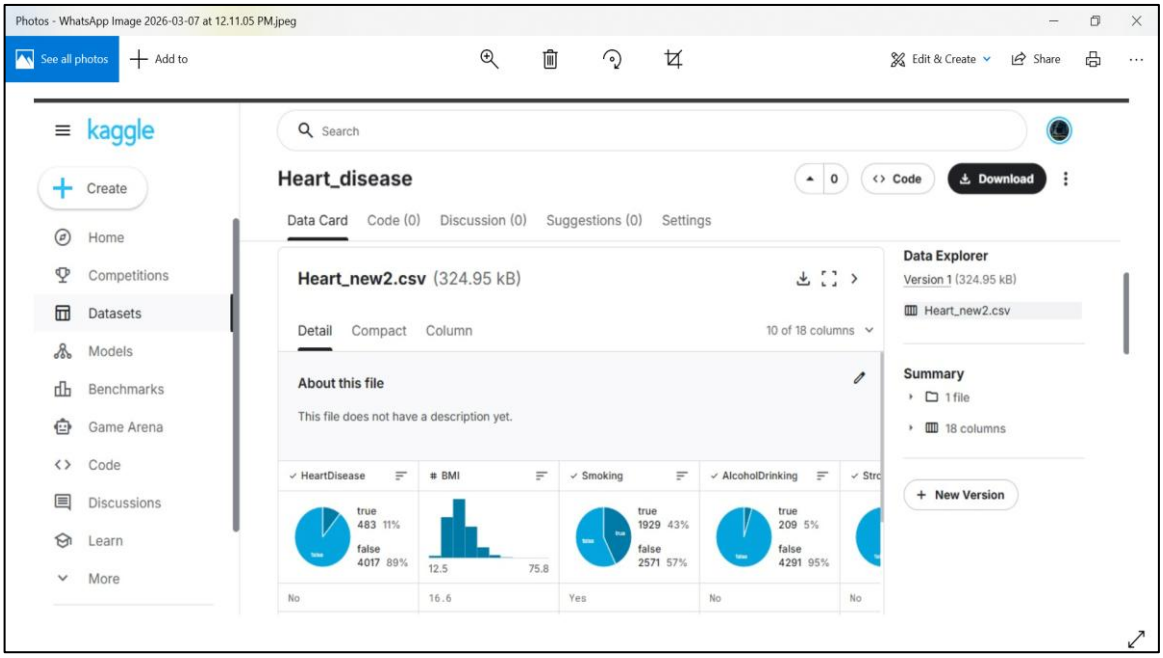
1) Data Collection & Extraction from Database:

✞ Collect the dataset:

1) Data is Collected from Dataset link Which is provided already so click on **Dataset Link**.



2) From the Kaggle Download the Dataset **Heart_new2.csv** by clicking on **Download** button,



✞ Loading the Dataset:

Step 1: Click on "Microsoft Excel" or "Text File"

On the Start Page, under Connect, you will see options like:

Microsoft Excel (for .xlsx)

Text File (for .csv)

Click the correct option.

Step 2: Select Your File

Browse your computer.

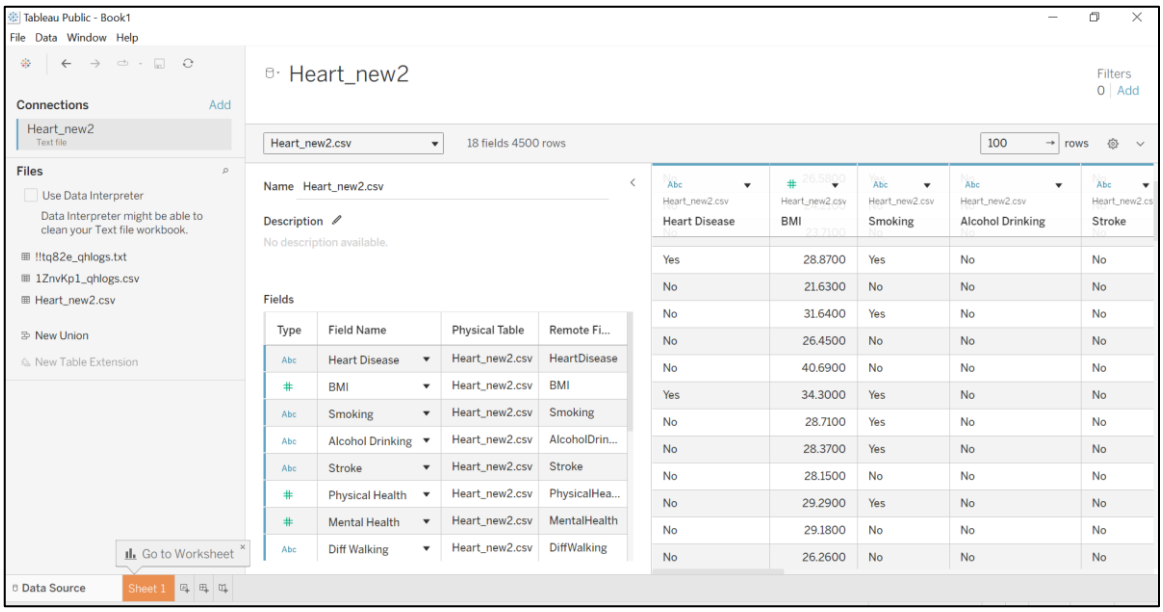
Select your dataset file. Click Open.

Step 3: Data Source Page

Your sheet names will appear on the left side.

Drag the sheet to the centre area.

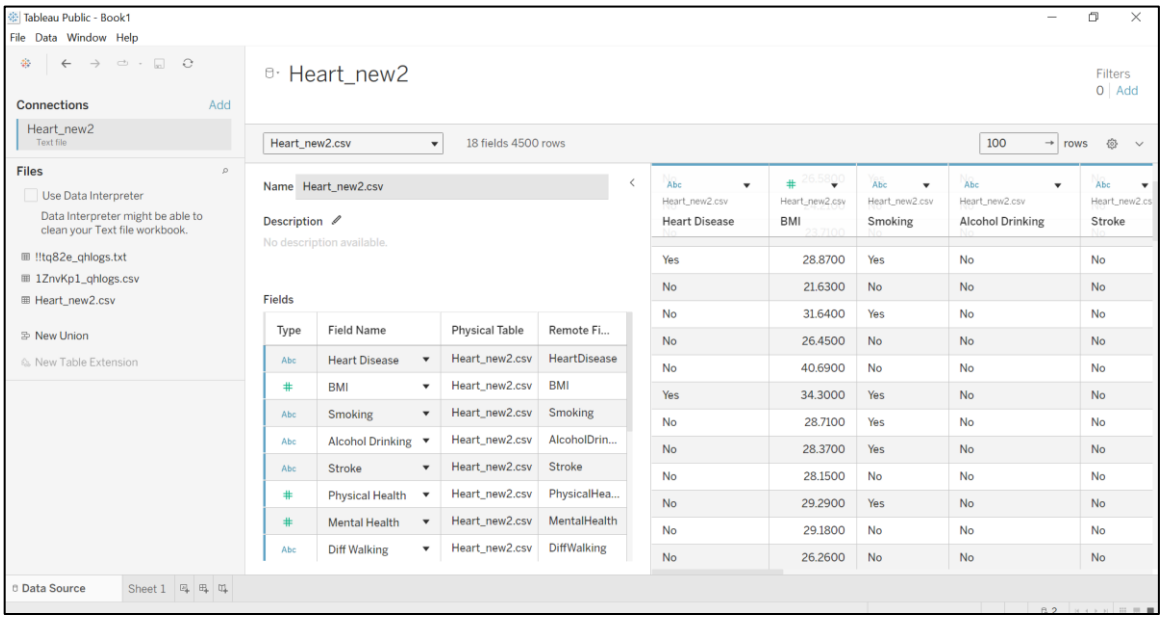
Tableau will show a preview of your data.



2) Data Preparation:

✞ Data Cleaning

Preparing the data for visualization involves cleaning the data to remove irrelevant or missing data, transforming the data into a format that can be easily visualized, exploring the data to identify patterns and trends, filtering the data to focus on specific subsets of data, preparing the data for visualization software, and ensuring the data is accurate and complete.



3) Data Visualization:

Data visualization is the process of creating graphical representations of data to help people understand and explore the information. The goal of data visualization is to make complex data sets more accessible, intuitive, and easier to interpret. By using visual elements such as charts, graphs, and maps, data visualizations can help people quickly identify patterns, trends, and outliers in the data.

✞ No of Unique Visualizations:

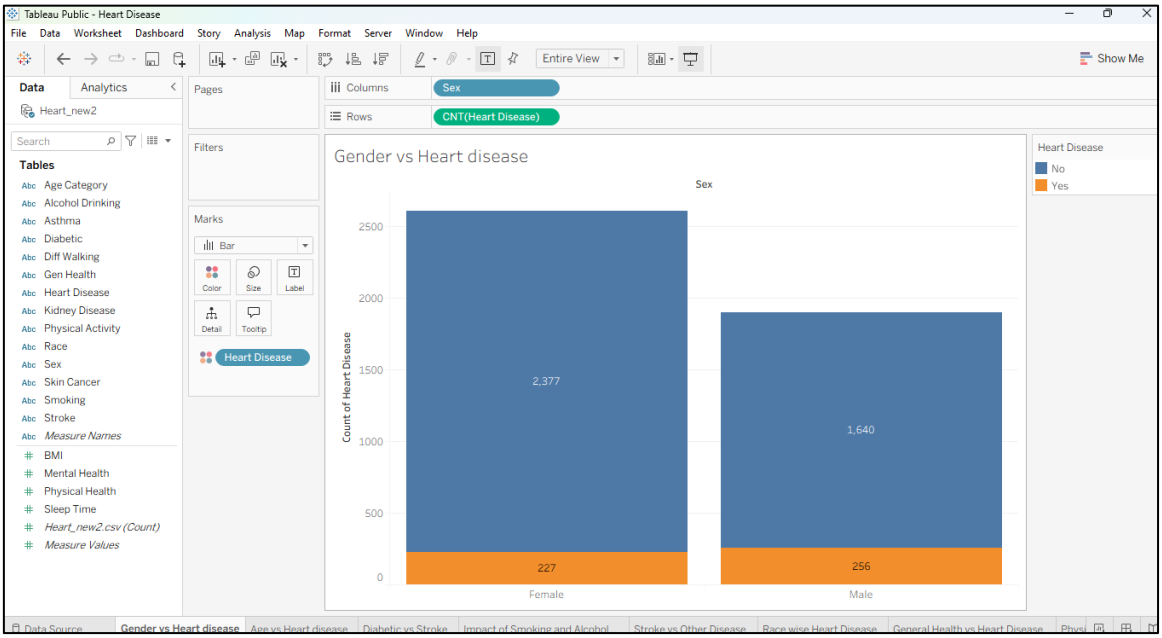
1) Gender vs. Heart Disease

a) Drag and Drop the Measure Name and Measure Values.

b) Add only-

- Count of Heart Disease
- Avg. Female & Male

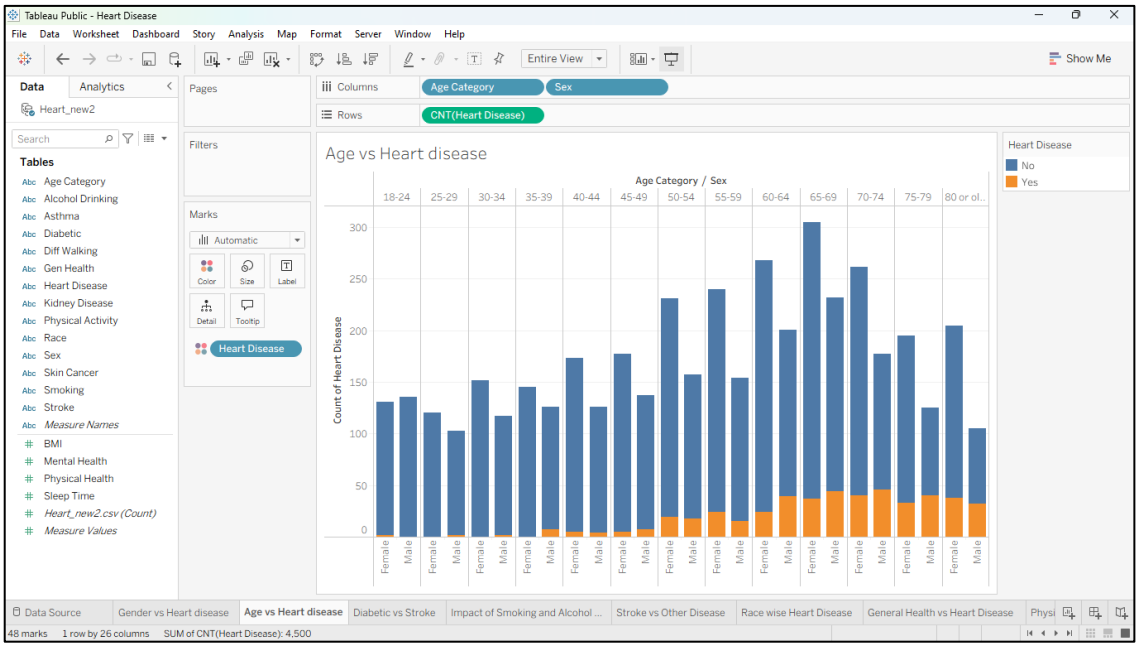
c) Drag and Drop Measure Name at Column section.



2) Age vs Heart Disease

a) Gender Differences Age

b) Drag and Drop Age category for grouping count of Heart Disease into column section, and rows.

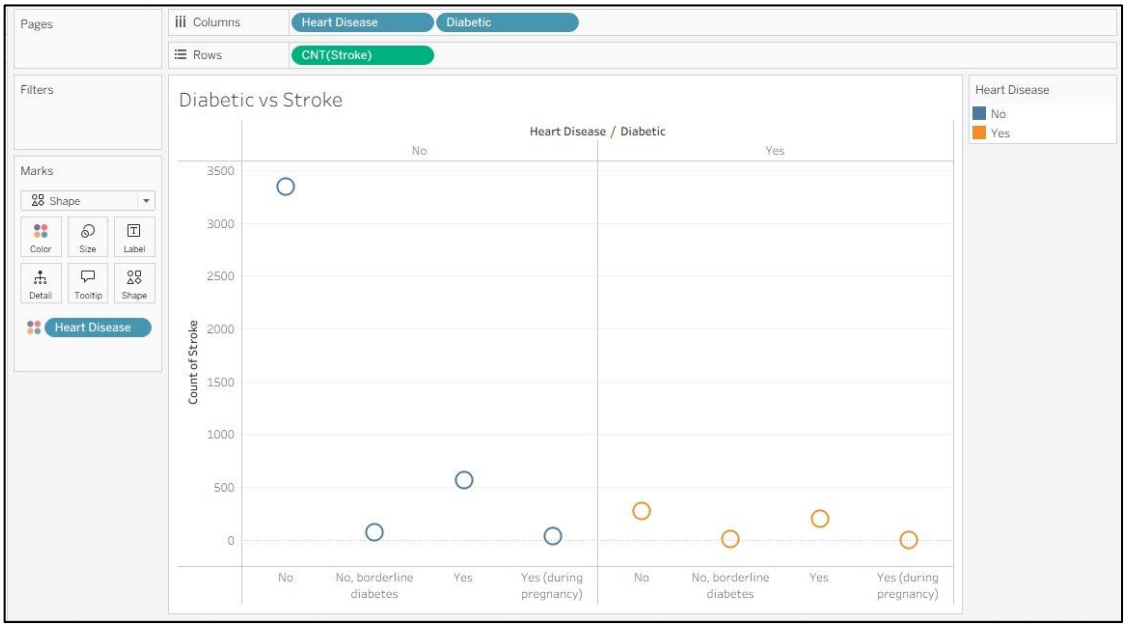


3) Dibeatic vs Stroke

a) The Heart Disease Factor.

b) Diabetic Status Breakdown.

c) Observation on Risk Correlation.



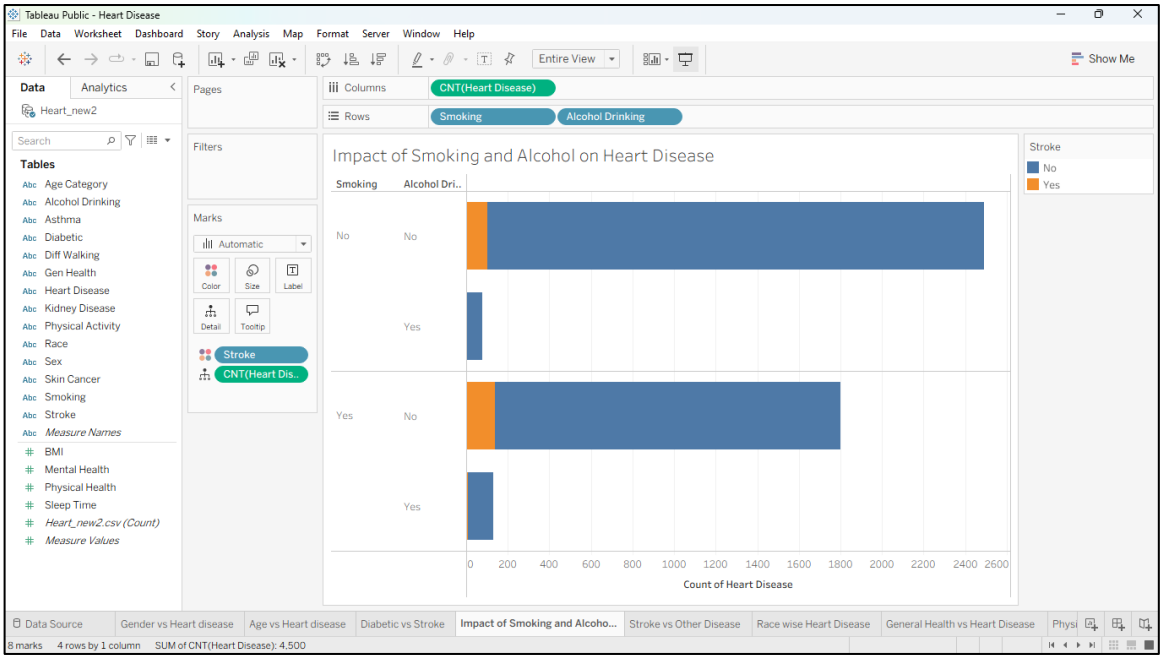
4) Impact of smoke and Alcohol on Heart Disease

1. Volume of Cases by Lifestyle Habit

a) Non-Smokers & Non-Drinkers: This is the largest group in the dataset, with a count exceeding 2,500 individuals.

b) Smokers & Non-Drinkers: This is the second-largest group, totaling approximately 1,800 individuals.

2. Stroke Correlation (The Orange Segments) The orange segments (Stroke = Yes) are visible across all lifestyle categories



5) Other Heart Disease

a) Highest Frequency Group.
b) Smoking Impact.
c) Alcohol Consumption.
d) Stroke Correlation

6) Racewise Heart Disease

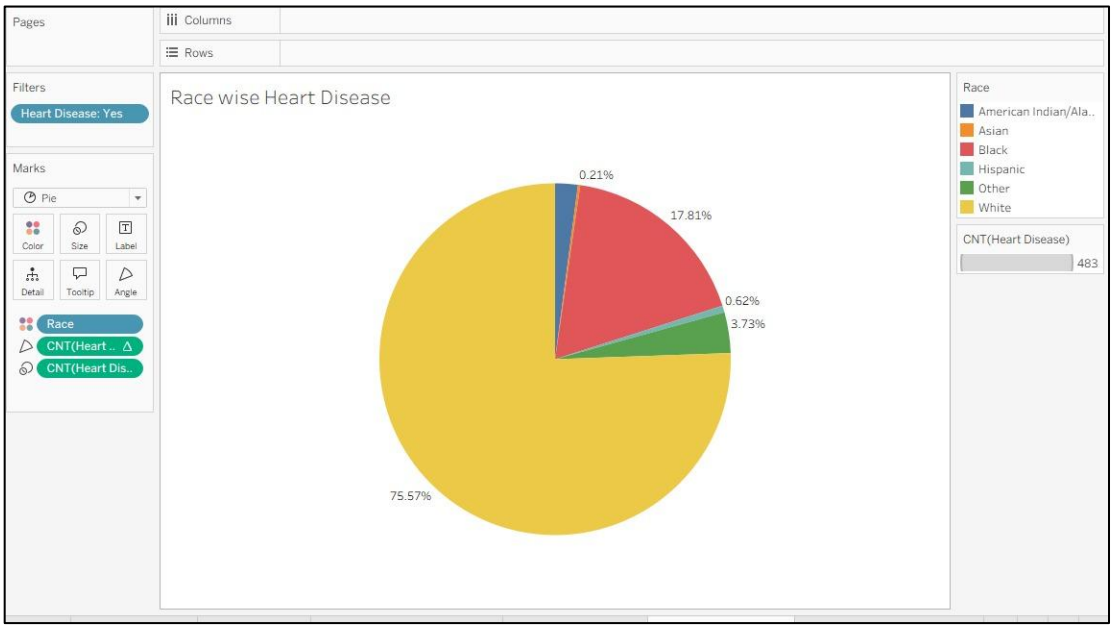
Largest Group: Individuals who neither smoke nor drink represent the largest portion of the dataset (over 2,400 cases).

Smoking Influence: Those who smoke but do not drink form the second largest group (approx. 1,800 cases).

Stroke Distribution: Stroke cases (orange) are clearly visible in both major categories but appear proportionately significant in the smoking group.

Total Sample: The status bar indicates a total count of 4,500 individuals for this analysis.

Race-wise Heart



7) General Health

Health Category Counts:

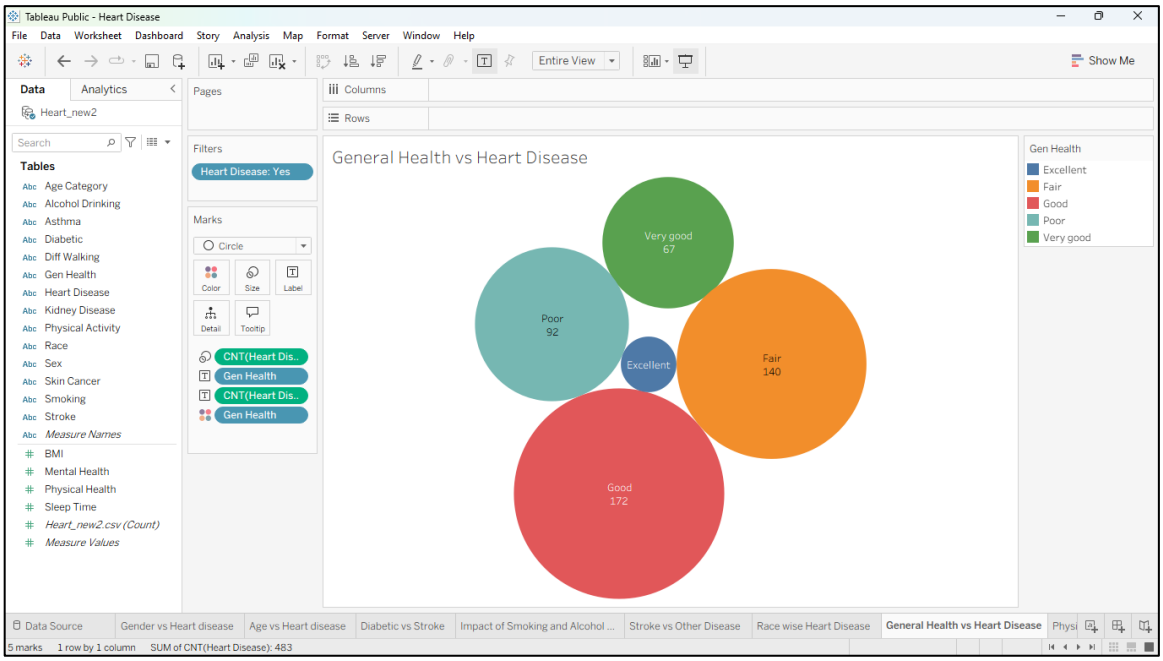
Good: 172 cases (the largest bubble).

Fair: 140 cases.

Poor: 92 cases.

Very good: 67 cases.

Excellent: This category has the smallest representation among heart disease patients.



8) Physical Activity

Demographic Breakdown:

White: 75.57%.

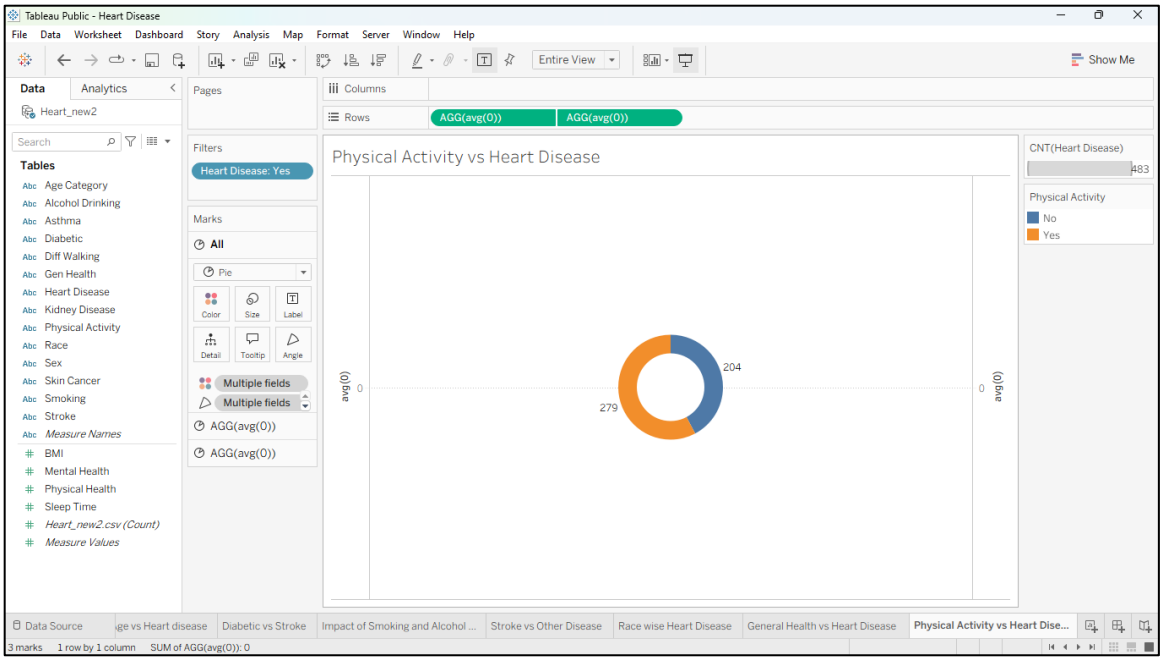
Black: 17.81%.

Other: 3.73%.

Hispanic: 0.62%.

Asian: 0.21%.

Total Volume: There are 483 total records represented in this specific demographic slice.



9) Age and BMI vs Diabetic

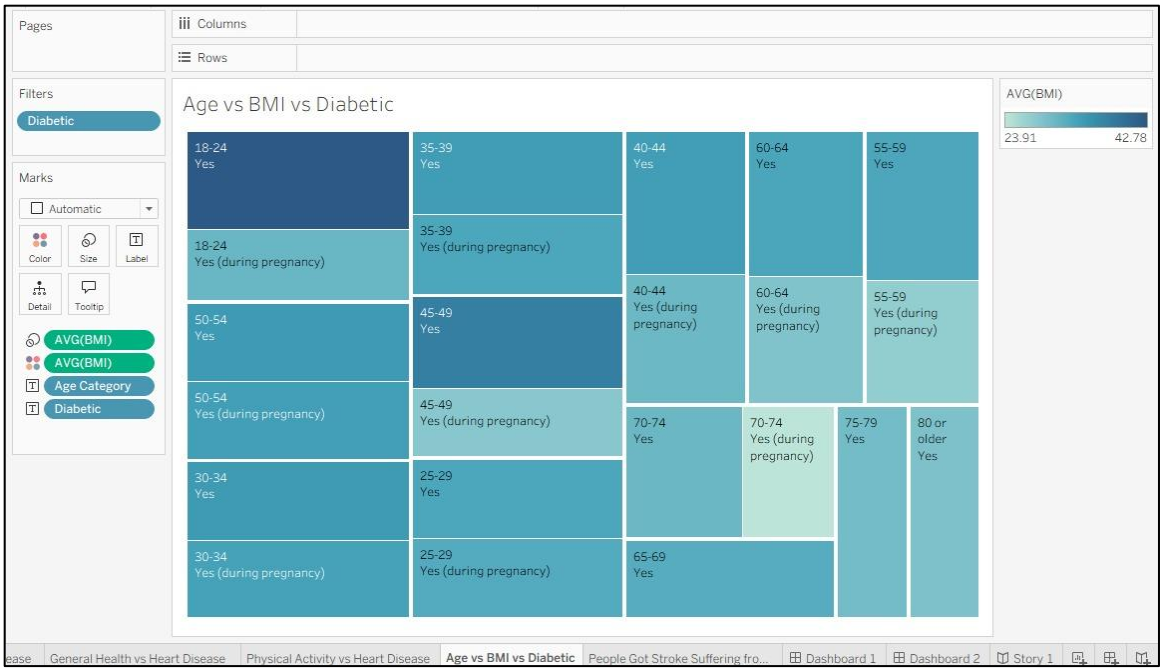
Age vs BMI vs Diabetic

Chart Type: Treemap.

Analysis: Compares Average BMI across different age categories and diabetic statuses.

Color Gradient: Darker blue indicates a higher Average BMI (up to 42.78), while lighter teal indicates a lower Average BMI (down to 23.91).

Observation: The 18-24 (Diabetic: Yes) group shows the highest average BMI intensity in this dataset.



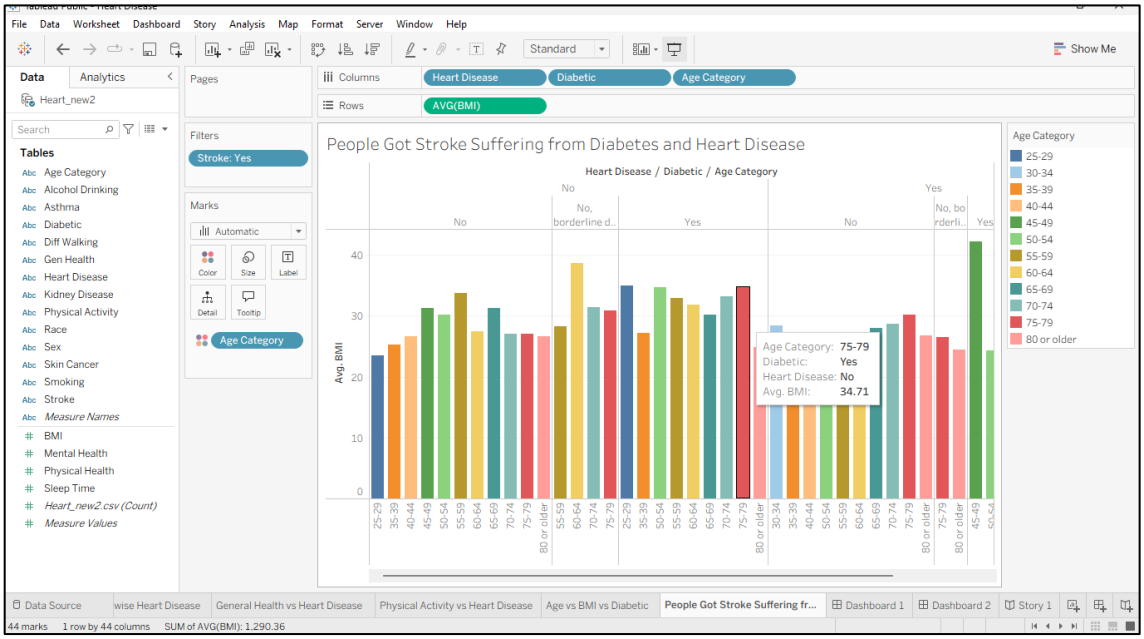
10) People Got Stroke Suffering from Heart Disease and Diabetic

Visualization Type: A Grouped Bar Chart analyzing BMI across multiple health conditions.

Filter Context: Specifically filters for individuals who have had a Stroke (Yes).

Columns: Categorized by Heart Disease (No/Yes), Diabetic (No/Borderline/Yes), and Age Category.

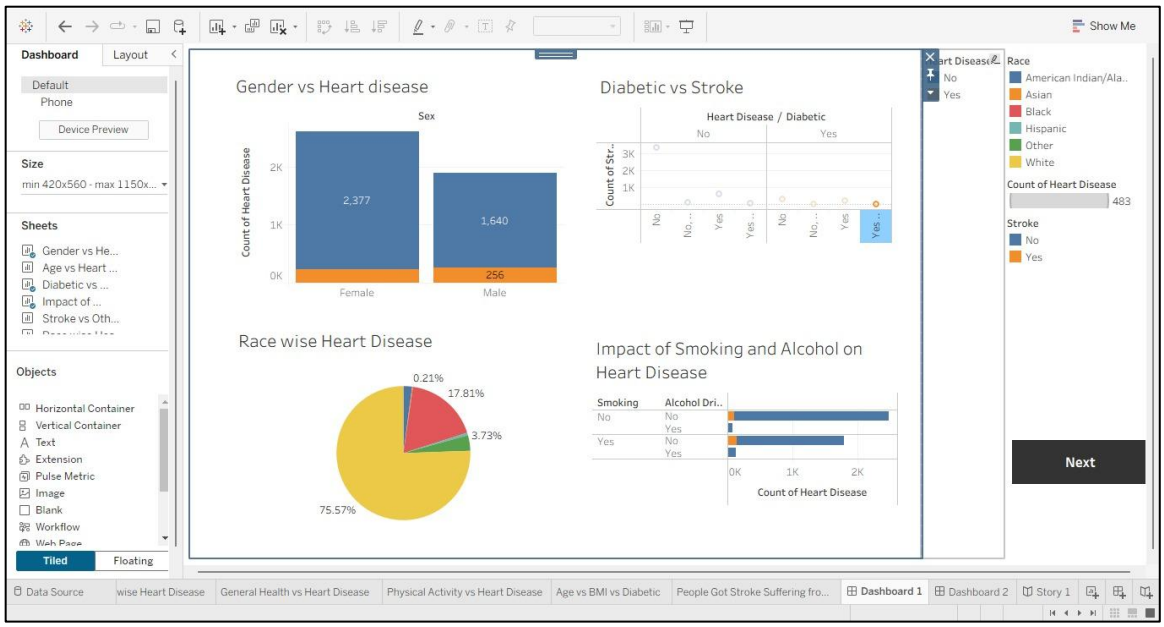
Highlighted Data: A tooltip shows that for individuals in the 75-79 age group who are Diabetic but have No Heart Disease, the Average BMI is 34.7

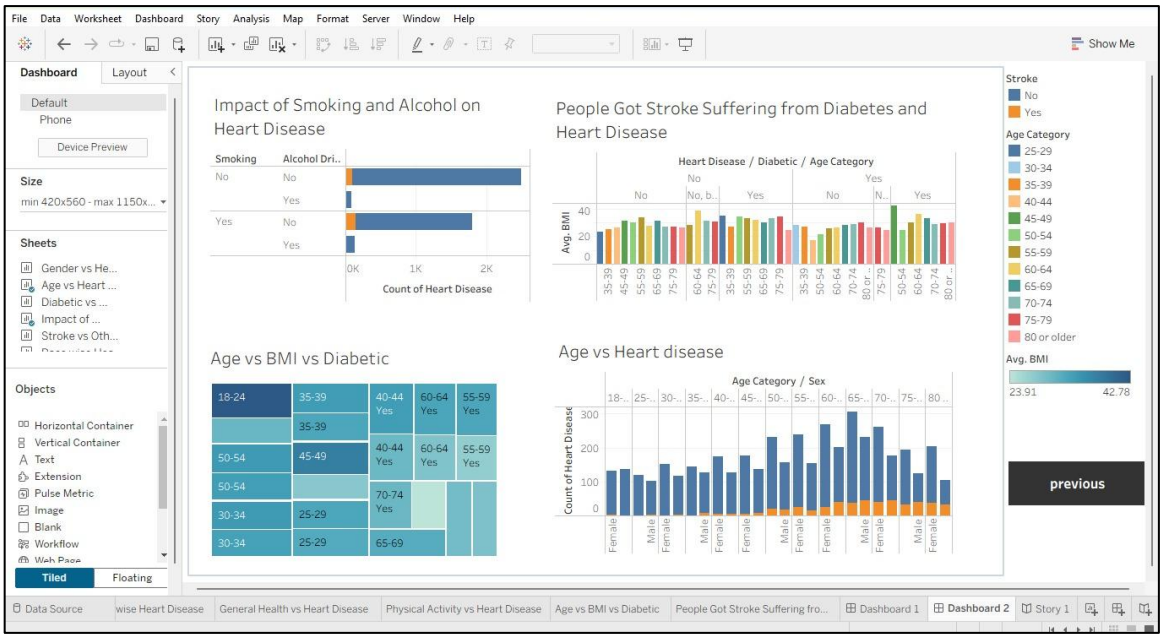


4) Dashboard

A dashboard is a graphical user interface (GUI) that displays information and data in an organized, easy-to-read format. Dashboards are often used to provide real-time monitoring and analysis of data and are typically designed for a specific purpose or use case. Dashboards can be used in a variety of settings, such as business, finance, manufacturing, healthcare, and many other industries. They can be used to track key performance indicators (KPIs), monitor performance metrics, and display data in the form of charts, graphs, and tables.

✞ Responsive and Design of Dashboard

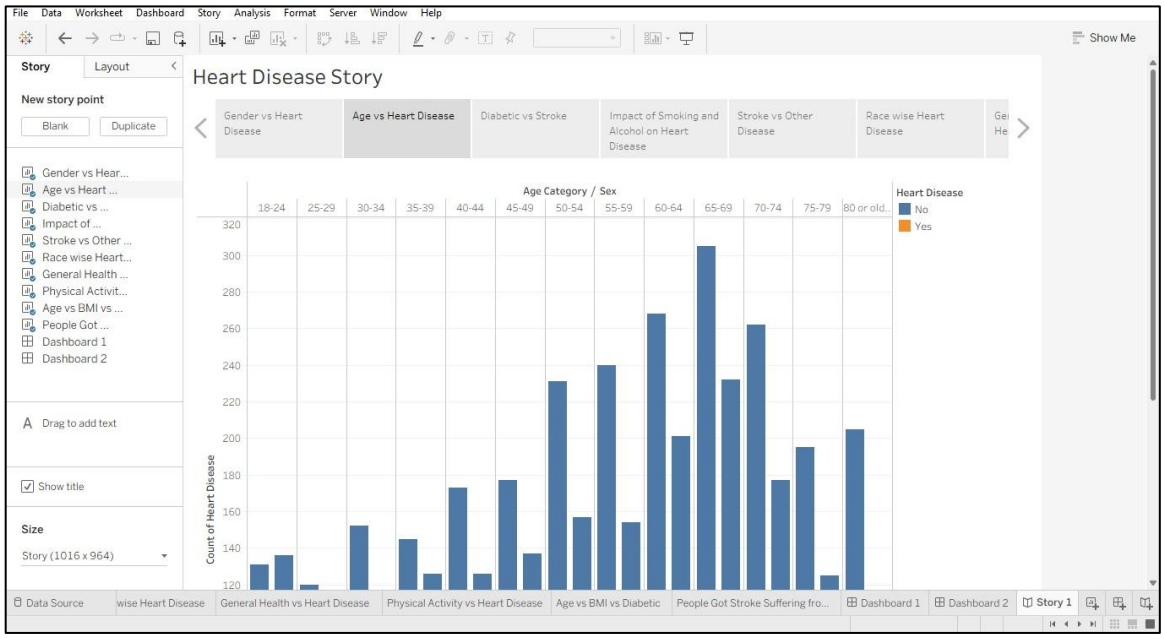




5) Storyboard

A storyboard is a visual representation of a sequence of events, typically used in multimedia projects such as films, animations, advertisements, or presentations. It consists of a series of drawings or images arranged in a sequence, often accompanied by annotations or descriptions, to outline the flow of the story or concept.

✞ Design a Storyboard



6) Performance Testing

Performance testing involves assessing the volume of data rendered from the database, the impact of data filters on system responsiveness, and the complexity introduced by the number of visualizations. Optimizing these factors ensures the dashboard operates efficiently, providing timely and reliable insights.

✞ Amount of Data Loaded

"Amount of Data Loaded" refers to the quantity or volume of data that has been imported, retrieved, or loaded into a system, software application, database, or any other data storage or processing environment. It's a measure of how much data has been successfully processed and made available for analysis, manipulation, or use within the system.

Columns Indexes Triggers Foreign Keys Partitions Grants DDL

Local instance MySQL80

heart_d.heart_new2

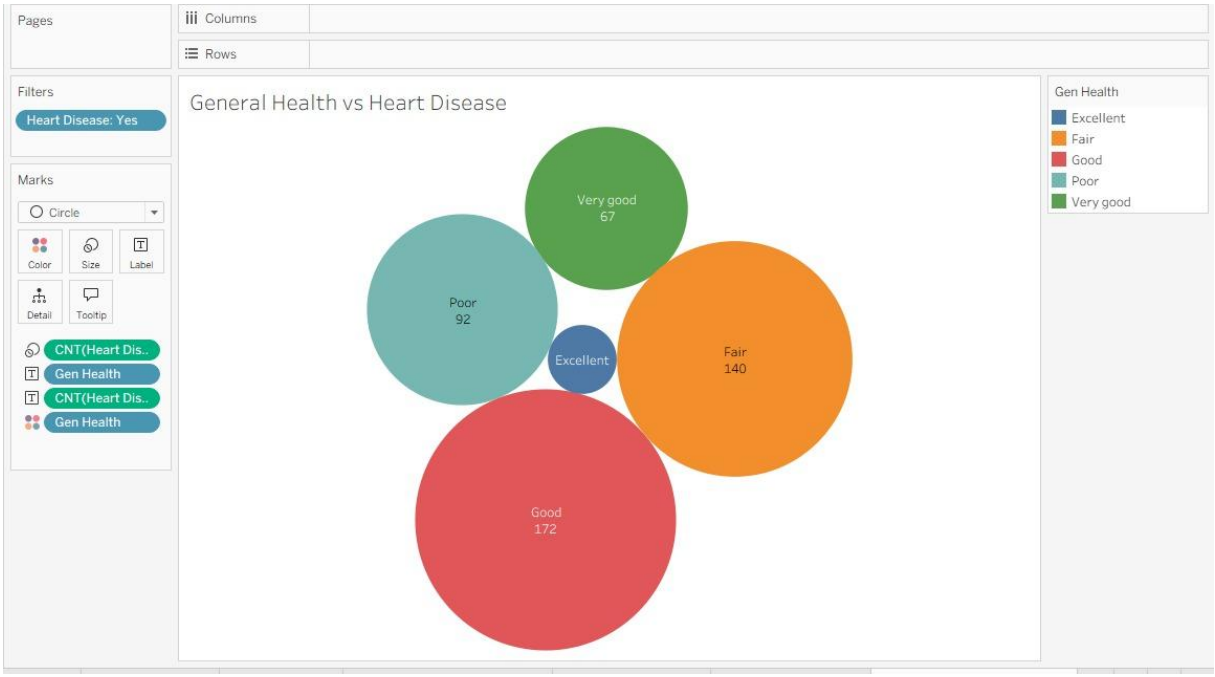
Table Details

Engine:	**InnoDB**
Row format:	**Dynamic**
Column count:	**18**
Table rows:	**4327**
AVG row length:	**367**
Data length:	**1.5 MiB**
Index length:	**0.0 bytes**
Max data length:	**0.0 bytes**
Data free:	**0.0 bytes**
Table size (estimate):	**1.5 MiB**
File format:	
Data path:	**C:\ProgramData\MySQL\MySQL Server 8.0\Data\heart_d\heart_new2.ibd**
Update time:	
Create time:	**2026-03-04 20:19:36**
Auto increment:	
Table collation:	**utf8mb4_0900_ai_ci**
Create options:	
Comment:	

Information on this page may be outdated. Click Analyze Table to update it.

✞ Utilization of Filters

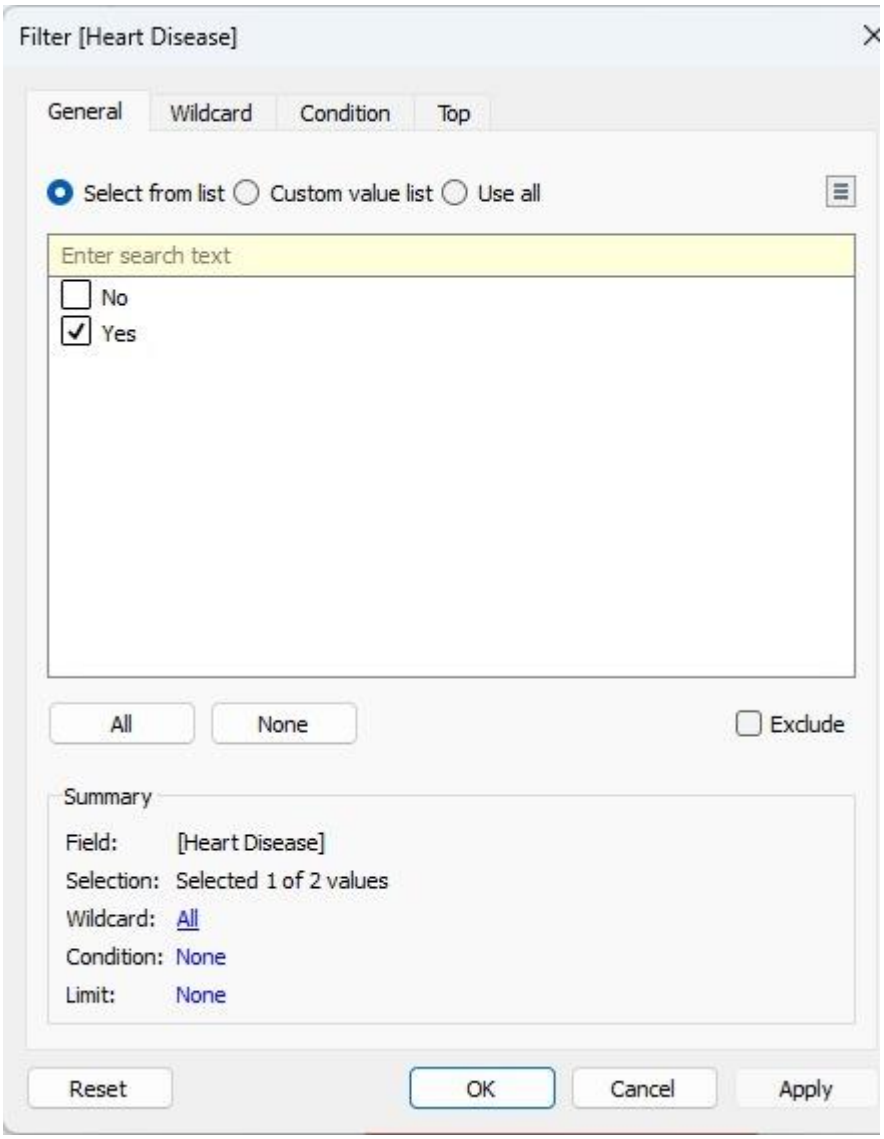
"Utilization of Filters" refers to the application or use of filters within a system, software application, or data processing pipeline to selectively extract, manipulate, or analyse data based on specified criteria or conditions.

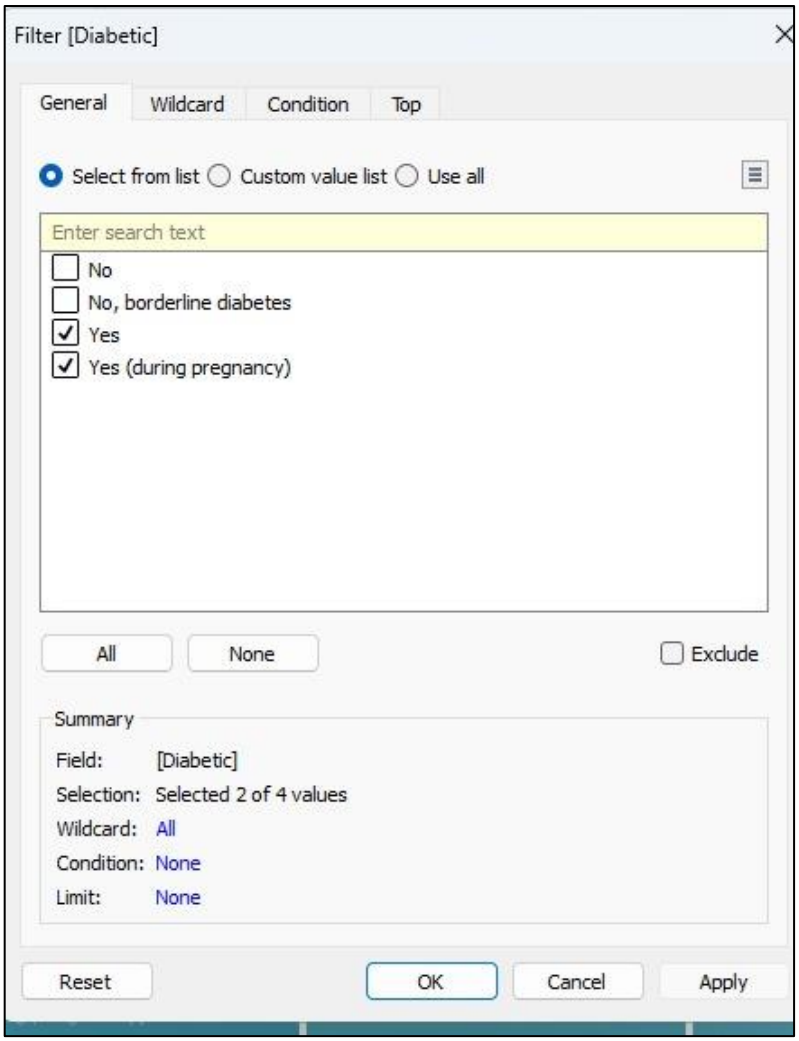


✞ No of Calculation Field

In **Tableau**, a **Calculated Field** is a custom field you create using formulas to perform computations on your data. It allows you to go beyond the raw fields in your dataset and generate new metrics or dimensions

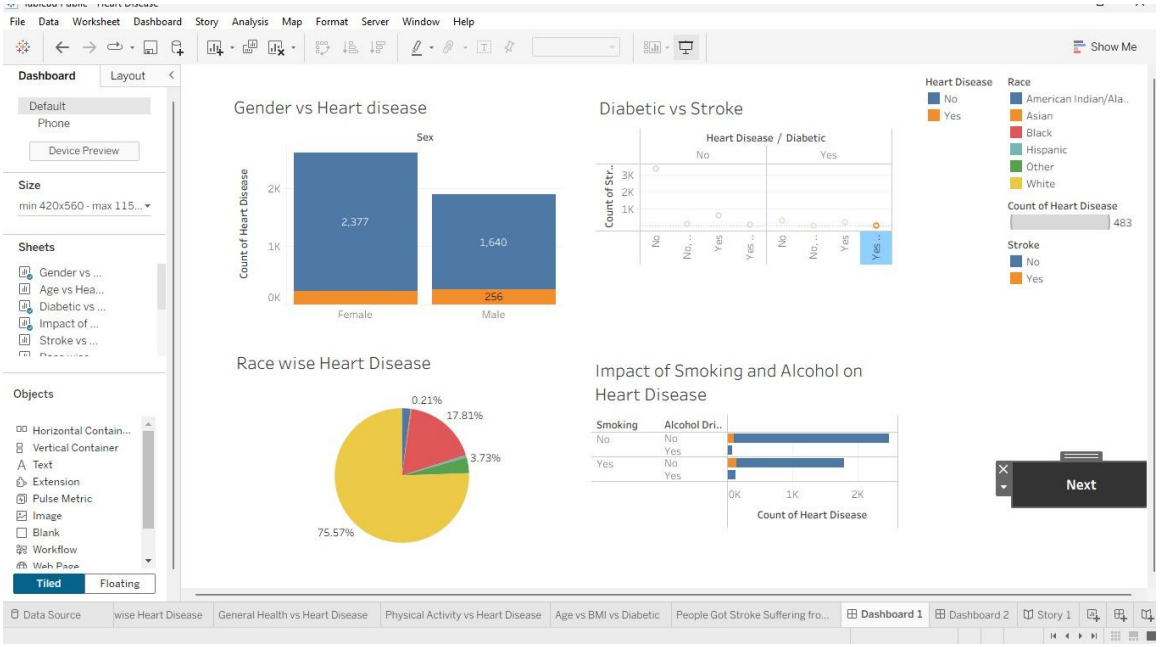
a) Right-clicking on the data pane and selecting **Create → Calculated Field**.
b) Giving it a name..
c) Writing your formula (e.g., IF [Cholesterol] > 200 THEN "High" ELSE "Normal" END).
d) Clicking **OK**, after which the new field appears in your data pane.




Filter [Diabetic]
General
Wildcard
Condition
Top
Select from list
Custom value list
Use all
Enter search text
No
No, borderline diabetes
Yes
Yes (during pregnancy)
All
None
Exclude
Summary
Field: [Diabetic]
Selection: Selected 2 of 4 values
Wildcard: All
Condition: None
Limit: None
Reset
OK
Cancel
Apply

❖ No of Visualizations/ Graphs

1) Count of Transformed Housing Data



7) Web integration

Web integration of a Tableau involves embedding interactive visualizations into a website or web application. This allows users to explore data insights directly within a web interface, enhancing accessibility and engagement. It supports real-time updates, user filtering, and seamless navigation for a dynamic data storytelling experience.

✞ Publishing

a) **Sign in to Tableau Public**

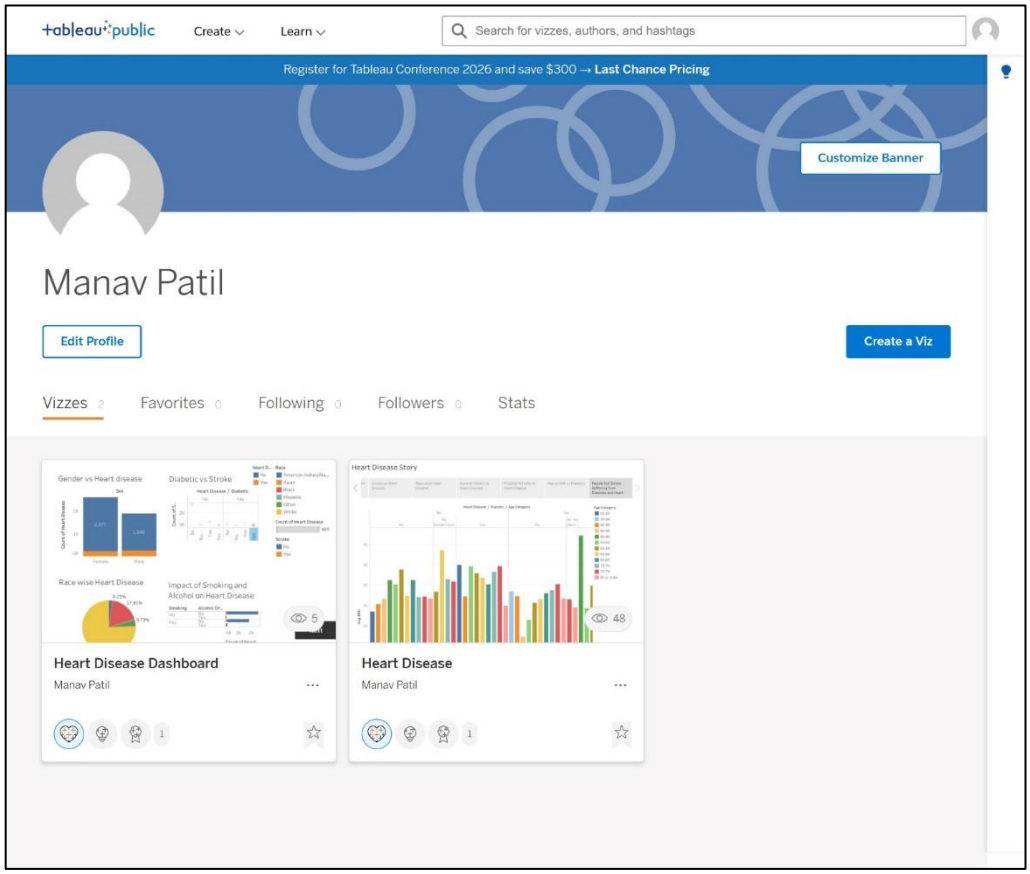
In Tableau Desktop, go to **File** > **Save to Tableau Public**.

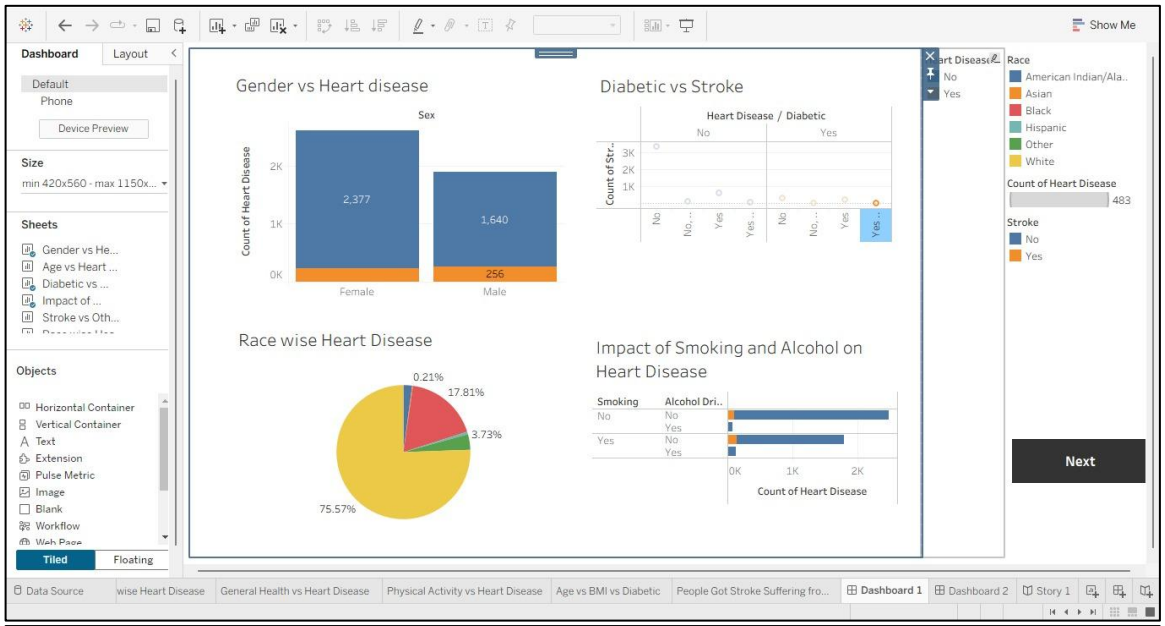
If you're not already signed in, a login window will appear.

Enter your **Tableau Public credentials** or sign up if you don't have an account.

b) **Save and Publish**

- After logging in, you'll be prompted to **name your workbook**.
- Click **Save** – Tableau will upload the workbook to your Tableau Public profile.




Gender vs Heart disease
Sex
Count of Heart Disease
2,377
1,640
256
Female
Male
Diabetic vs Stroke
Heart Disease / Diabetic
Race wise Heart Disease
0.21%
17.81%
3.73%
75.57%
Impact of Smoking and Alcohol on Heart Disease
Smoking
Alcohol Dri..
Count of Heart Disease
Race
American Indian/Ala..
Asian
Black
Hispanic
Other
White
Count of Heart Disease
483
Stroke
No
Yes
Next
Dashboard
Layout
Sheets
Objects
Tiled
Floating
Data Source
Dashboard 1
Dashboard 2
Story 1

Pages
Columns
Heart Disease
Diabetic
Age Category
Rows
AVG(BMI)
Filters
Stroke: Yes
Marks
Automatic
Color
Size
Label
Detail
Tooltip
Age Category
People Got Stroke Suffering from Diabetes and Heart Disease
Heart Disease / Diabetic / Age Category
Avg. BMI
Age Category
25-29
30-34
35-39
40-44
45-49
50-54
55-59
60-64
65-69
70-74
75-79
80 or older

✞ Dashboard and Story embed with UI with Flask.

In this activity, you will create a simple web application using Flask and embed a Tableau Dashboard or Story into the UI. This allows users to view and interact with Tableau visualizations directly through a custom web interface.

1) app.py

```python
import pickle
from flask import Flask, render_template, request

app= Flask(__name__)

@app.route('/')
def home():
    return render_template('index.html')

if __name__== '__main__':
    app.run(debug=False)
```

2) Adding Dashboard and Story on HTML page.

a) Open Tableau Public profile and open Dashboard/Story.

b) Click on share option and copy the embed code.

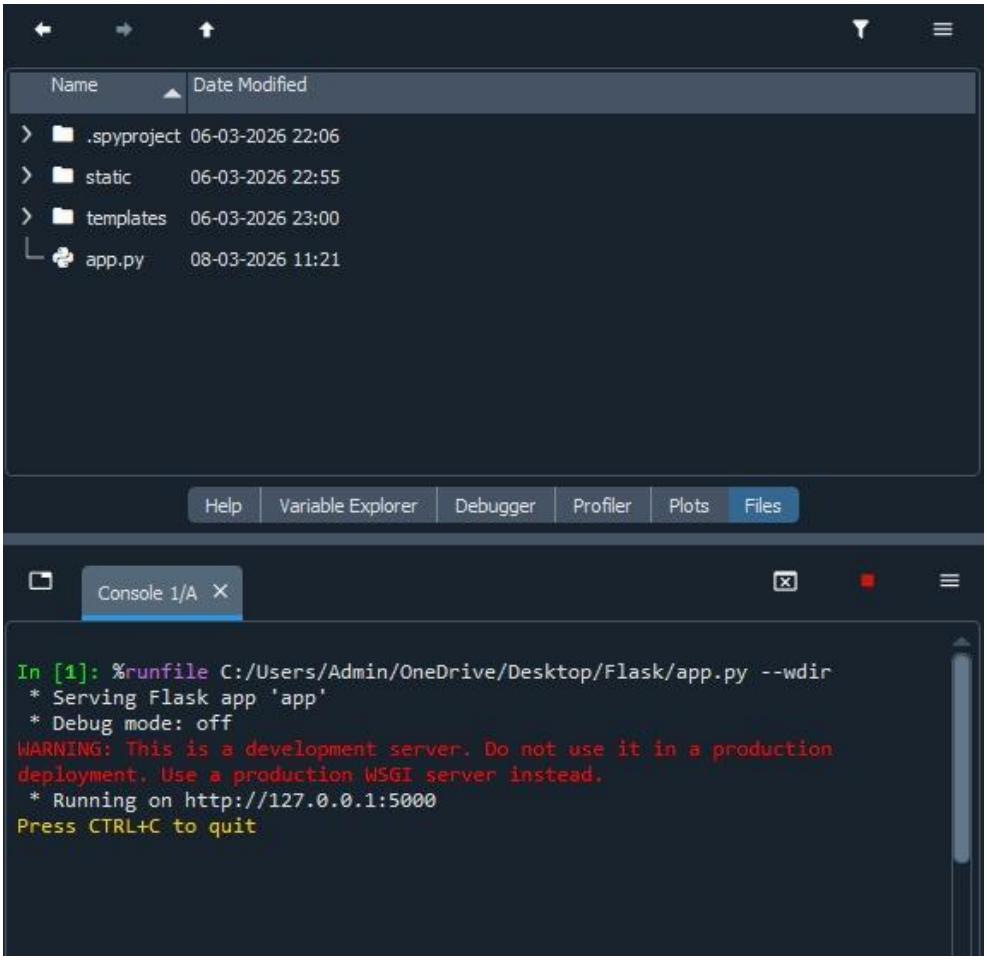
c) Paste in HTML page

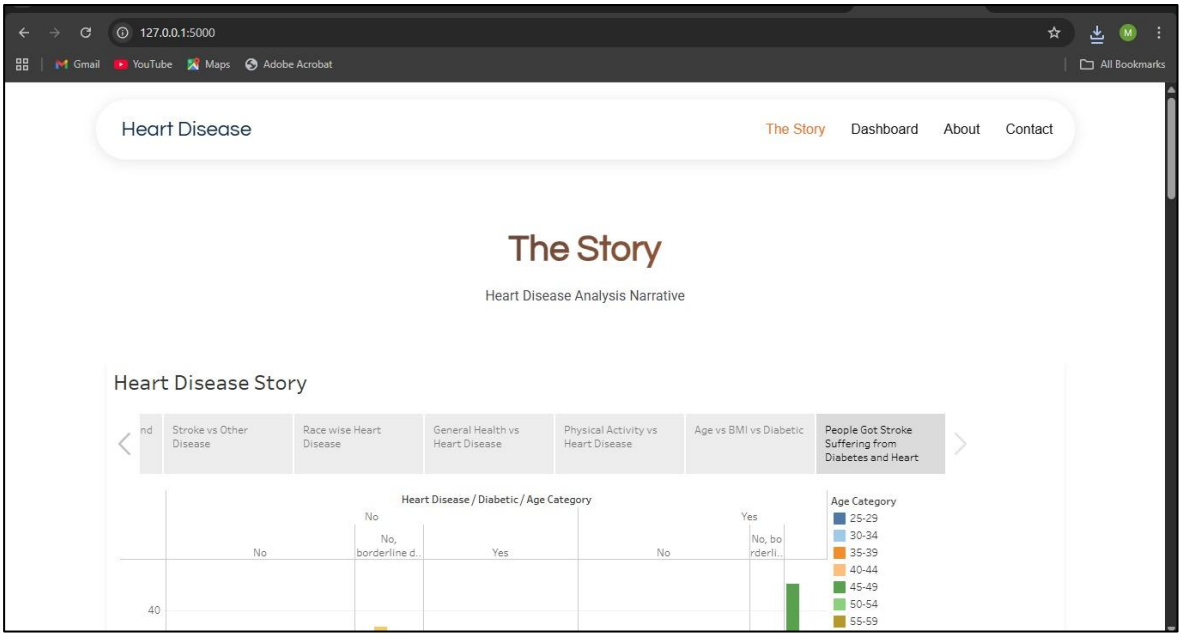
C:\Users\Admin\OneDrive\Desktop\Flask\templates\index.html

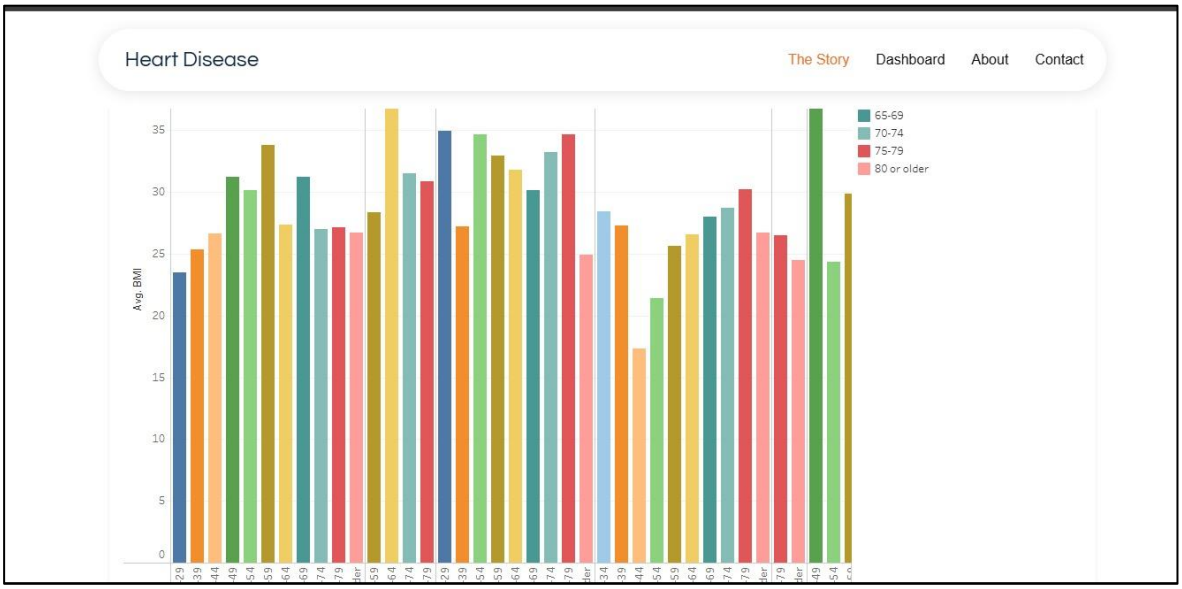
app.py × index.html ×

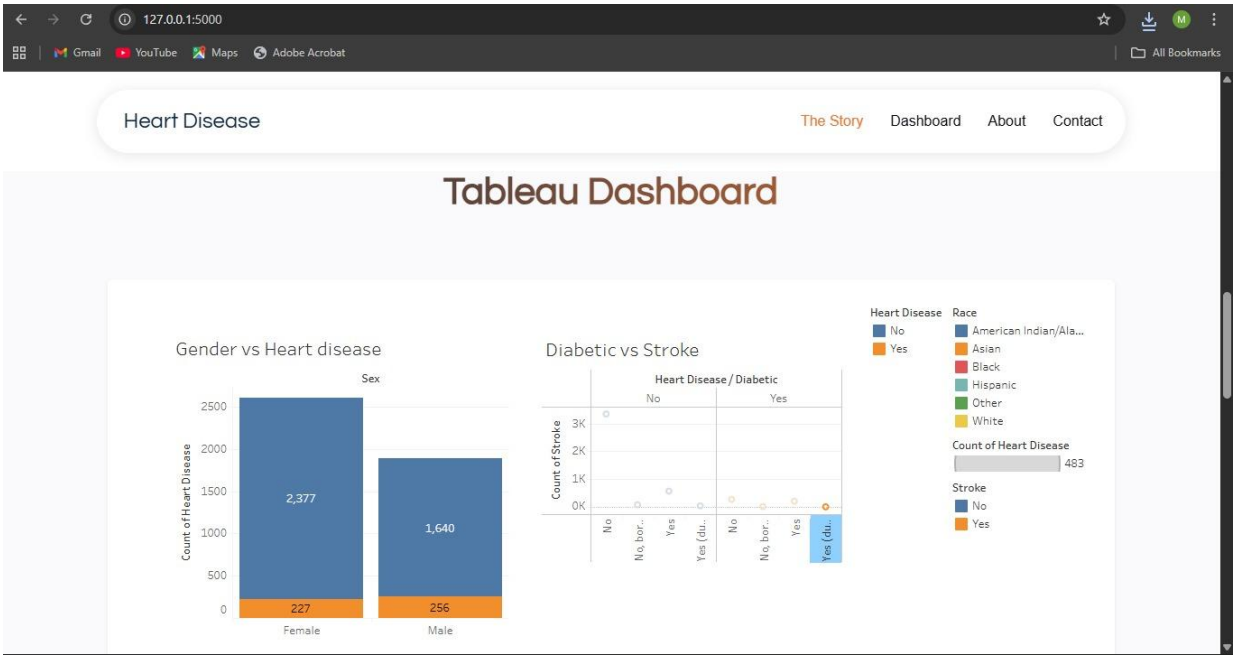
```html
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="utf-8">
  <meta content="width=device-width, initial-scale=1.0" name="viewport">
  <title>Data Portfolio - Heart Disease Analysis</title>

  <link href="{{ url_for('static', filename='img/favicon.png') }}" rel="icon">

  <link href="https://fonts.googleapis.com" rel="preconnect">
  <link href="https://fonts.gstatic.com" rel="preconnect" crossorigin>
  <link href="https://fonts.googleapis.com/css2?family=Roboto:wght@300;400;700&family=Questrial&display=swa

  <link href="{{ url_for('static', filename='vendor/bootstrap/css/bootstrap.min.css') }}" rel="stylesheet">
  <link href="{{ url_for('static', filename='vendor/bootstrap-icons/bootstrap-icons.css') }}" rel="styleshe
  <link href="{{ url_for('static', filename='vendor/aos/aos.css') }}" rel="stylesheet">

  <link href="{{ url_for('static', filename='css/main.css') }}" rel="stylesheet">
</head>

<body class="index-page">

  <header id="header" class="header d-flex align-items-center sticky-top">
    <div class="header-container container-fluid container-xl position-relative d-flex align-items-center j
      <a href="/" class="logo d-flex align-items-center me-auto">
        <h1 class="sitename">MyPortfolio</h1>
      </a>
      <nav id="navmenu" class="navmenu">
        <ul>
          <li><a href="#story" class="active">The Story</a></li>
          <li><a href="#dashboard">Dashboard</a></li>
          <li><a href="#about">About</a></li>
          <li><a href="#contact">Contact</a></li>
        </ul>
      </nav>
```

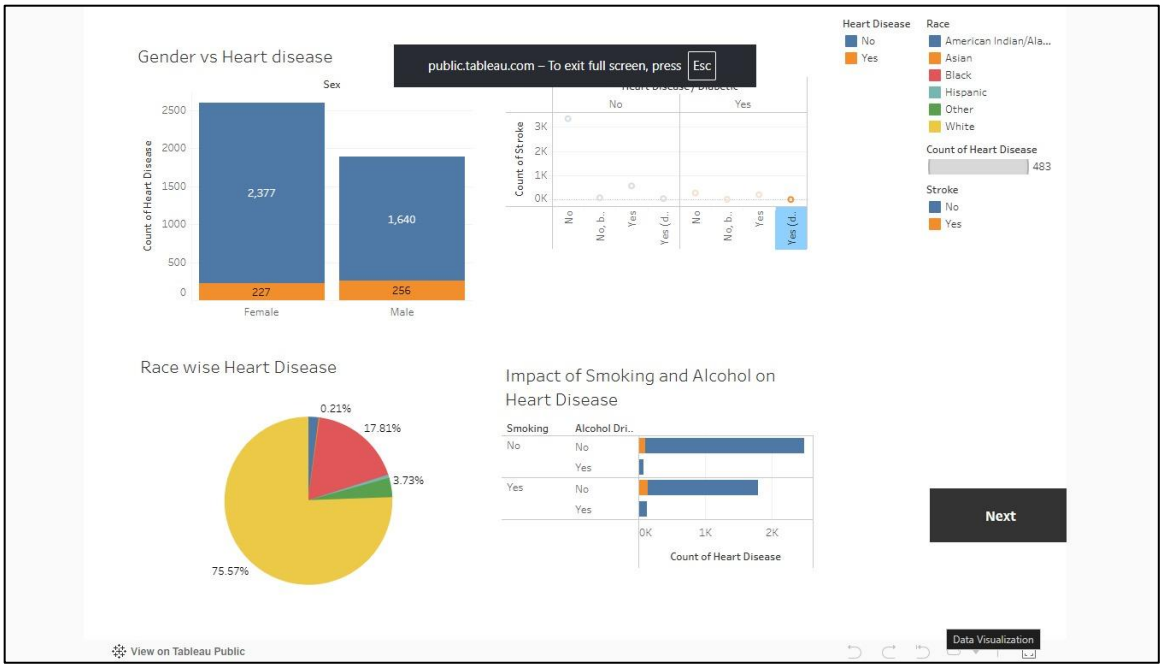
d) Run the app.py and copy the port number and paste it on Browser.


Name
Date Modified
.spyproject 06-03-2026 22:06
static 06-03-2026 22:55
templates 06-03-2026 23:00
app.py 08-03-2026 11:21
Help
Variable Explorer
Debugger
Profiler
Plots
Files
Console 1/A
In [1]: %runfile C:/Users/Admin/OneDrive/Desktop/Flask/app.py --wdir
* Serving Flask app 'app'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit


127.0.0.1:5000
Gmail
YouTube
Maps
Adobe Acrobat
All Bookmarks
Heart Disease
The Story
Dashboard
About
Contact
The Story
Heart Disease Analysis Narrative
Heart Disease Story
Stroke vs Other Disease
Race wise Heart Disease
General Health vs Heart Disease
Physical Activity vs Heart Disease
Age vs BMI vs Diabetic
People Got Stroke Suffering from Diabetes and Heart
Heart Disease / Diabetic / Age Category
Age Category
25-29
30-34
35-39
40-44
45-49
50-54
55-59


Heart Disease
The Story
Dashboard
About
Contact
Avg. BMI
65-69
70-74
75-79
80 or older


127.0.0.1:5000
Gmail
YouTube
Maps
Adobe Acrobat
All Bookmarks
Heart Disease
The Story
Dashboard
About
Contact
Tableau Dashboard
Gender vs Heart disease
Sex
Count of Heart Disease
2,377
1,640
227
256
Female
Male
Diabetic vs Stroke
Heart Disease / Diabetic
No
Yes
Count of Stroke
Heart Disease
No
Yes
Race
American Indian/Ala...
Asian
Black
Hispanic
Other
White
Count of Heart Disease
483
Stroke
No
Yes


Gender vs Heart disease
public.tableau.com – To exit full screen, press Esc
Sex
Count of Heart Disease
2500
2000
1500
1000
500
0
2,377
1,640
227
256
Female
Male
Heart Disease
No
Yes
Race
American Indian/Ala...
Asian
Black
Hispanic
Other
White
Count of Heart Disease
483
Stroke
No
Yes
Count of Stroke
3K
2K
1K
0K
No
Yes
Race wise Heart Disease
0.21%
17.81%
3.73%
75.57%
Impact of Smoking and Alcohol on Heart Disease
Smoking
Alcohol Dri..
No
Yes
0K
1K
2K
Count of Heart Disease
Next
View on Tableau Public
Data Visualization